MODULE 5.2

Firewalls in Network Design

Security Architectures | Layered Defense

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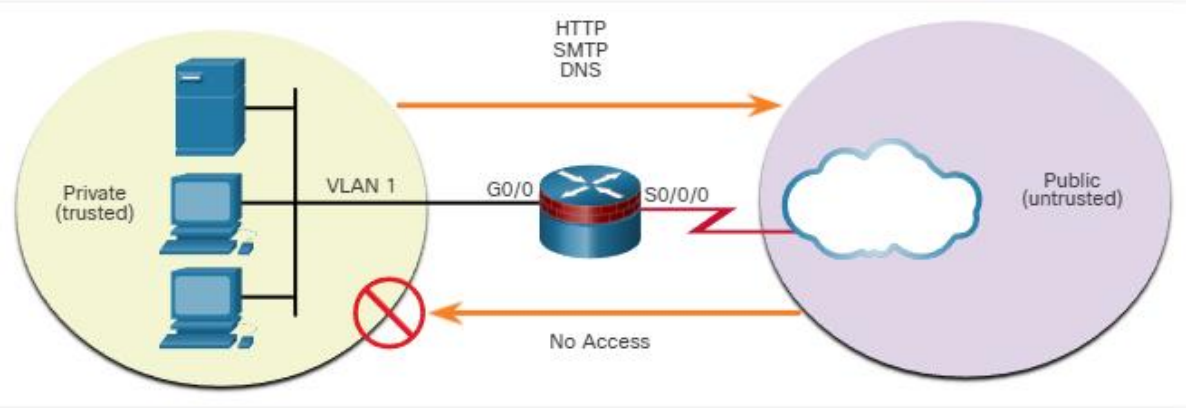
Common Security Architectures: Private and Public

- **Two-interface design:** the simplest firewall architecture designates just two zones — an outside (public, untrusted) network and an inside (private, trusted) network, each tied to a specific firewall interface.

- **Private → public:** traffic originating from the private network is permitted and inspected as it travels toward the public network; inspected return traffic belonging to that same conversation is permitted back in.

- **Public → private:** traffic originating from the public network and heading toward the private network is generally blocked outright — there's no reason for the outside to initiate contact with the inside.

- **This topology:** hosts on VLAN 1 (private/trusted) reach HTTP, SMTP, and DNS services out on the public network through R1; nothing from the public side can reach back in uninvited.



Abbreviations: VLAN = Virtual LAN. HTTP = HyperText Transfer Protocol. SMTP = Simple Mail Transfer Protocol. DNS = Domain Name System.

Common Security Architectures: Demilitarized Zone (DMZ)

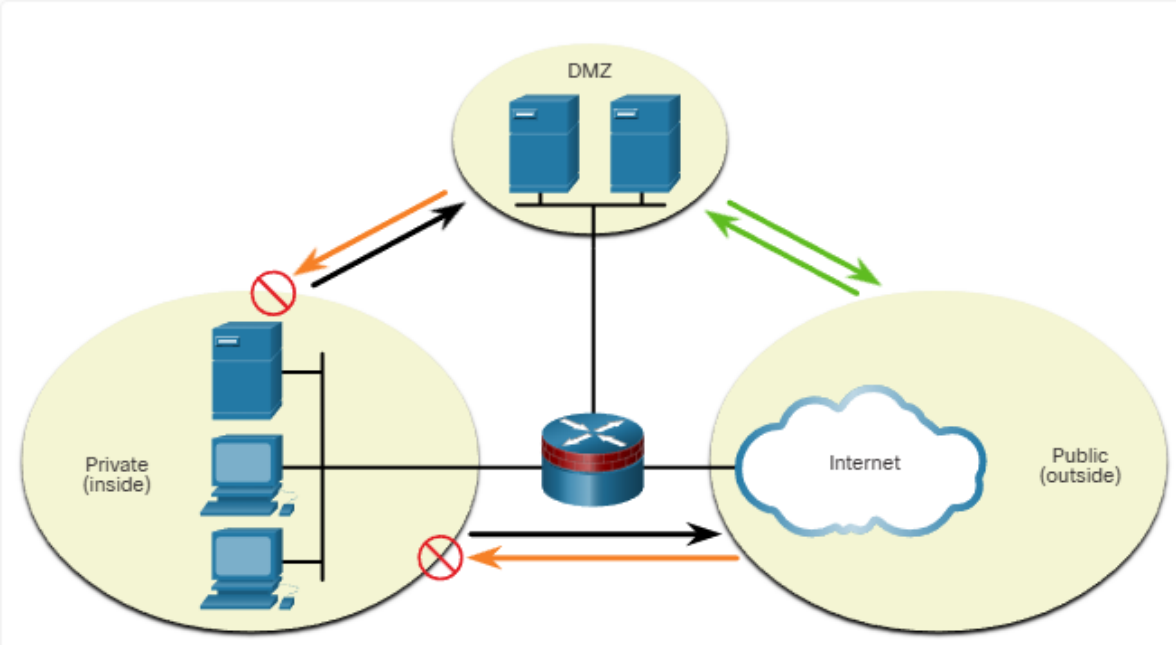
- **Three-interface design:** a DMZ adds a third zone — servers that need to be reachable from the public internet (web, mail, DNS) sit in their own zone, isolated from the fully trusted private network.

- **Private → DMZ/public:** inspected and permitted with little restriction; matching return traffic is allowed back.

- **DMZ → private:** blocked by default — even though DMZ servers are semi-trusted, they must never reach into the private network directly. This protects the inside if a DMZ server gets compromised.

- **Public → DMZ:** selectively permitted and inspected, limited to the specific services the DMZ actually offers; return traffic is dynamically permitted.

- **Public → private:** still fully blocked, same as the two-interface design.



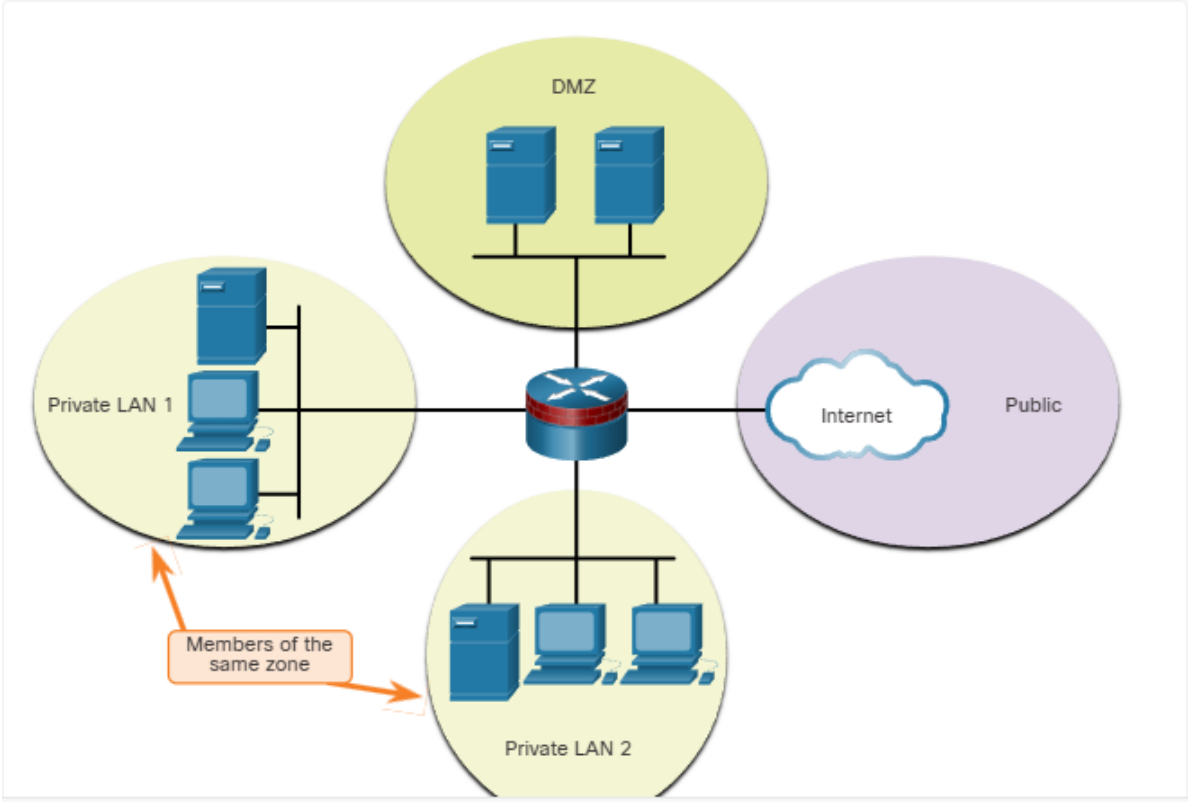
Common Security Architectures: Zone-Based Policy Firewalls

- **A zone:** a group of one or more interfaces that share similar security requirements — e.g., Private LAN 1 and Private LAN 2 can be grouped into the same zone if they need the same policy.

- **Same-zone traffic:** passes freely by default — no policy is applied between interfaces inside one zone.

- **Zone-to-zone traffic:** blocked by default (a "deny any" posture) — a policy must be explicitly configured to permit or inspect traffic crossing between zones.

- **The self zone:** the one exception — it represents the router itself and all its interface IP addresses, relevant to management and control-plane traffic like SSH, SNMP, and routing protocols. There's no default policy for it either.



Layered Defense

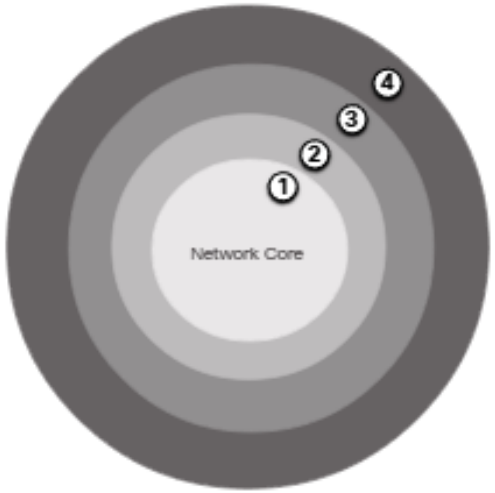
- **Defense in depth:** instead of relying on one firewall at the perimeter, a layered design wraps protection around the network in concentric rings, so a single failure doesn't expose everything behind it.

- **Network core security** (innermost ring) — protects against malicious software and traffic anomalies, enforces network policies, and ensures survivability of the most critical assets.

- **Perimeter security:** secures the boundaries between zones — this is where most traditional firewall rules live.

- **Communications security:** provides information assurance — confidentiality and integrity for data as it moves.

- **Endpoint security** (outermost ring) — provides identity and device security policy compliance, making sure only trusted, compliant devices even get to participate.



1. ***Network Core security*** *- Protects against malicious software and traffic anomalies, enforces network policies, and ensures survivability*
2. ***Perimeter security*** *- Secures boundaries between zones*
3. ***Communications security*** *- Provides information assurance*
4. ***Endpoint security*** *- Provides identity and device security policy compliance*

Check Your Understanding

Read the design description, then name the network security design concept it matches:

Q: A firewall with only two interfaces — one tagged "trusted," one "untrusted" — with no middle ground for public-facing servers.
A: Private and Public Architecture

Q: Web, mail, and DNS servers sit in their own isolated zone so a compromise there can't reach the fully trusted internal network.
A: Demilitarized Zone (DMZ)

Q: Two LANs with identical security requirements are grouped so traffic between them passes freely, while traffic to any other group is denied by default.
A: Zone-Based Policy Firewall (same zone)

Q: Security wraps the network in rings — core, perimeter, communications, and endpoint — so no single layer is a single point of failure.
A: Layered Defense

Summary

- Every firewall design boils down to one rule: traffic initiated from the trusted side is generally permitted and inspected; traffic initiated from the untrusted side is denied unless explicitly configured otherwise.

- **5.2.1 —** three common architectures: Private/Public (two interfaces, simplest), DMZ (three interfaces, isolates public-facing servers from the trusted network), and Zone-Based Policy Firewalls (groups interfaces into zones, deny-any between zones by default).

- **5.2.2 —** layered defense wraps protection in rings — network core, perimeter, communications, and endpoint security — so no single control is a single point of failure.

- **5.2.3 —** recognizing which architecture or layer a described behaviour belongs to is the real skill — the underlying deny-by-default logic is shared across all of them.

References

Cisco Networking Academy — Network Defense, Module 5.2: Firewalls in Network Design

https://www.netacad.com

Cisco — Zone-Based Policy Firewall Design Guide

https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/98628-zone-design-guide.html

Cisco — DMZ Design Guidelines

https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/23602-confaccesslists.html